TRANSLATION HANKOOK

PORSCHE Inspection Catalog Chassis: Tires

Testing Procedure F.360.104
Spring characteristic and static dimensions

Test Carrier:	Test Rig
Testing Facility:	Test Rig of Tire Manufacturer
Status (Porsche):	10.07.2019
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(see target data sheet / ”Zieldatenblatt”)

1 Purpose

The purpose of the test procedure is to verify the static dimensions and to record general technical data of tires.

2 Field of Application

This test procedure establishes criteria after which the required technical data for the approval of tire specifications are determined. The test procedure can be used at any stage of the product life and compared with the approval status of the respective tire specification.

3 Definition of Terms

Operating point = wheel load according to target data sheet

4 Responsibility

Approval test:
The test for approval is performed and documented by the tire manufacturer (self-certification). The results are made available to the PAG immediately.

5 Description of the Testing Procedure

5.1 Measuring Equipment

5.1.1 Measured Variables / Assessment Variables

Measured variable	**Measurement tolerance**
Wheel load	±0.5 %
Spring deflection	±0.5 %
Tire weight	±0.05 kg
Shore hardness of the tread	±2
Bead to bead distance	±5 mm
Max. tire width 2.0/2.7/3.4 bar (balanced)	±1 mm
Tire diameter 2.0/2.7/3.4 bar (balanced)	±1 mm
Tread depth of all tread grooves (containing TWI)	±0.5 mm
Tire envelope (dimensioned)	±1 mm
Tread pattern of tire contact patch under wheel load from target data sheet (dimensioned)	±1 mm

5.1.2 Installation of the transducer

N/A

5.1.3 Data Processing

N/A

5.2 Test Conditions

The static deflection of the tire standing on flat ground shall be measured step by step with increasing wheel load with the following parameters:

- Ambient temperature should be 20° C - 25° C

5.3 Test Carrier / Test Set-Up

- Wheel load according to target data sheet = operating point no. 5 (of the data sheet Spring Characteristic Stat. Dim.)
  Start of measurement = operating point minus 2,000 N (-4 steps with step size - 500 N)
  End of measurement = working point plus 2,500 N (+5 steps for increment + 500 N)
- Camber = 0
- Reifendruck mit 2.0 bar, 2.7 bar, 3.4 bar
- Test rim according to target data sheet

5.4 Test Procedure

Measure the static deflection of the tire standing on flat ground step by step with increasing wheel load with increment +500N/-500N (see 5.3 - Start and end of measurement).

6 Documentation

6.1 Data Analysis

N/A

6.2 Presentation of the results

For each test specification presented, the Porsche Test Catalog - Data Sheet must be prepared, containing the following characteristics:

- Tire dimension, specification number, DOT number
- Tire weight
- Shore hardness of the tread
- Outer dimension of the distance between the two beads
- Tire diameter balanced at the respective air pressure 2.0 bar, 2.7 bar and 3.4 bar
- maximum tire width balanced at the respective air pressure 2.0 bar, 2.7 bar and 3.4 bar
- Tread depth of all tread grooves (containing TWI)
- Envelope curve **dimensioned** with tire pressure from target data sheet (according to sample template from data sheet Spring characteristic stat. dim.)
- Contact area **dimensioned** with tire pressure and wheel load from target data sheet

The static axle base distance and the radial deformation of the tire must be entered in the respective table as the result over the wheel load at the three required air pressures 2.0 bar, 2.7 bar and 3.4 bar. The radial stiffness of the tire is calculated.

The template for this is, according to the figure below, the data sheet ”Spring characteristic stat. dim.” based on the current Porsche test catalog.

FIGURE MISSING, CHECK ORIGINAL DOCUMENT

6.3 Requirements

The geometric dimensions of the tires must meet the requirements from the target data sheet or from the technical data or PLB.

7 Applicable Documents

Target data sheet
Porsche AG, data sheet: ”Spring characteristic and static dimensions F.360.104”
Technical Data
PLB (Porsche Performance Specification)

8 Distributor

EFF